Medical Imaging Dataset Exploration

Chest X-Ray NIH (https://pmc.ncbi.nlm.nih.gov/articles/PMC10607847/)

The NIH Chest X-ray dataset is a foundational, massive multi-label benchmark, excellent for training robust backbones via transfer learning, though it has text-mined label noise and heavily skewed class distributions. The dataset consisted of 112,120 X-rays from the front of 32,717 unique patients. There were 86,524 and 25,596 samples in the training and test sets, respectively.

The dataset is split into 15 Classes: Infiltration, Effusion, Atelectasis, Nodule, Mass, Pneumothorax, Consolidation, Pleural Thickening, Cardiomegaly, Emphysema, Edema, Fibrosis, Pneumonia, Hernia, and Normal cases.

For the imbalances in the old dataset they are as follows:
Over 50% of the dataset contains "No Finding" (normal cases) , certain conditions like Infiltration (~17.7%) and Effusion (~11.8%) are heavily represented.
In contrast, critical conditions such as Pneumonia (~1.3%) and Hernia (~0.2%) have a critically low sample count

According to the website itself :
Several concerns have been observed with the official split technique, in which the training and test datasets had different characteristics. This could be either due to a tremendous inconsistency in the label or the test set having an average of 3 times more photos per patient compared to the training set. Because of that, instead of using an official split, a custom split was used The new split was better balanced, had lower divergence, and did not use some images from single patients significantly more often than others.

Ekansh Jain 25b0651

Skin Lesion Kaggle (https://www.kaggle.com/datasets/andrewmvd/isic-2019)
(https://www.researchgate.net/figure/Overview-of-ISIC-2019-Dataset-and-The-Number-of-Images-in-Each-Classes_tbl1_365804093)

The ISIC 2019 dataset provides a realistic look into dermatological computer-aided diagnostics. It forces us to handle class imbalances and real-world image properties like lens lighting and hair, making it an exceptional testing ground for advanced color constancy, segmentation, and lesion preprocessing techniques.

The dataset for ISIC 2019 contains 25,331 images available for the classification of dermoscopic images among nine different diagnostic categories:

- Melanoma
- Melanocytic nevus
- Basal cell carcinoma
- Actinic keratosis

- Benign keratosis (solar lentigo / seborrheic keratosis / lichen planus-like keratosis)
- Dermatofibroma
- Vascular lesion
- Squamous cell carcinoma
- None of the above

Data imbalances:
Melanocytic Nevus (NV) is overwhelmingly the majority class, capturing over 12,000+ images (nearly 50% of the entire dataset).

Several Concerns are: The ISIC 2019 dataset was curated by combining multiple sub-datasets. Rare or early-stage lesions like Dermatofibroma (DF) and Vascular lesions (VASC) are extremely few in the datasets, dermoscopic images frequently contain extraneous artifacts (air bubbles, ink markings, rulers, calibration charts, etc.)

Ekansh Jain 25b0651